# Chemistry Topic 7 — Chemistry in Daily Life, Industry and Agriculture

**Sources baked in:** NCERT Class 10 & 12 (Chemistry in Everyday Life), ICAR Agriculture Chemistry Fundamentals, Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on Urea nitrogen percentage (46%), Crookes glass UV absorption, Silver Bromide in photography vs Silver Iodide in artificial rain, Sodium chlorate as herbicide, Zinc phosphide as rodenticide, DDT as non-biodegradable pollutant, and Dettol components. **Last verified:** September 2026

## Current Affairs (2024–2026)

| Year | Event | Core Concept & Must-Score Significance |
|---|---|---|
| **2025–2026** | **Nano Urea & Nano DAP Expansion** | IFFCO commercialization of **Nano Urea** ($4\%$ N w/v liquid nanoparticles, $20–50$ nm) and **Nano DAP**: Provides $> 80\%$ nutrient uptake efficiency via foliar stomatal absorption, eliminating massive nitrogen leaching into groundwater and cutting conventional bagged urea consumption. |
| **2024** | **UPPCS Prelims 2024** | Question on **Photography Salt vs Artificial Rain**: Highlighted the recurring pair: **Silver Bromide ($AgBr$)** is the light-sensitive photographic emulsion salt, whereas **Silver Iodide ($AgI$)** is the cloud-seeding agent for artificial rain. |
| **2023** | **UPPCS / UKPCS Trends** | Persistent testing of **DDT (Dichlorodiphenyltrichloroethane)**: Identified strictly as a **non-biodegradable organochlorine pollutant** that undergoes severe trophic biomagnification in food chains (thinning avian eggshells). |
| **2022** | **UPPCS Prelims 2022** | Re-tested **Crookes Glass**: Keyed as the specialty glass containing Cerium oxide ($CeO_2$) that sharply cuts off and **absorbs ultraviolet (UV) radiation** in sunglasses. Pyrex/Borosilicate was an incorrect distractor. |
| **2021** | **UPPCS Prelims 2021** | Agricultural Herbicide: Keyed **Sodium Chlorate ($NaClO_3$)** and **2,4-D** as non-selective and selective weedicides/herbicides, respectively. |

## Detailed Theory Notes

### 7.1 Cleansing Chemistry: Saponification, Micelles & Synthetic Surfactants

#### 1. Saponification Reaction

Soaps are manufactured by heating natural fats and oils (esters of glycerol with long-chain fatty acids) with aqueous alkali:

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CH₂ - O - CO - C₁₇H₃₅                        CH₂ - OH
|                                             |
CH  - O - CO - C₁₇H₃₅  +  3 NaOH  ——>        CH  - OH  +  3 C₁₇H₃₅COONa
|                                             |            (Sodium Stearate Soap)
CH₂ - O - CO - C₁₇H₃₅                        CH₂ - OH
(Glyceryl Stearate / Fat)                    (Glycerol)
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- The byproduct **Glycerol** is separated by fractional vacuum distillation and used in pharmaceuticals, cosmetics, and nitroglycerin explosives.

#### 2. Micelle Mechanics

- The hydrophobic hydrocarbon tail is lipophilic (attracted to fats, repelled by water).
- The hydrophilic carboxylate head is ionic and solvated by water molecules.
- When concentration exceeds the **Critical Micelle Concentration (CMC)**, molecules self-assemble into a spherical cluster of 100–200 molecules.
- The non-polar dirt/grease is entrapped within the hydrophobic interior core. The charged carboxylate heads on the outer shell repel other micelles, preventing coalescence and keeping the oil droplets suspended as an emulsion that washes away with water.

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### 7.2 Pharmaceutical Chemistry & Antiseptic Physics

#### 1. Analgesics & Antipyretics

- **Aspirin (Acetylsalicylic Acid)**:
- Synthesized by acetylating salicylic acid with acetic anhydride in the presence of sulphuric acid:

$$C_6H_4(OH)COOH + (CH_3CO)_2O \rightarrow C_6H_4(OCOCH_3)COOH + CH_3COOH$$

- Inhibits the enzyme cyclooxygenase (COX), blocking the biosynthesis of **prostaglandins** (pain and fever mediators) and **thromboxanes** (blood clotting agents).
- Used in low doses ($75–150\ \mathrm{mg/day}$) as an antiplatelet prophylactic to prevent coronary thrombosis and secondary strokes.

#### 2. Antiseptic vs. Disinfectant Chemistry

- **Dettol**: The active germicide is **Chloroxylenol (4-chloro-3,5-dimethylphenol)**, blended with $\alpha$-terpineol in pine oil and soap solution. It disrupts bacterial cell membranes without irritating human skin.
- **Phenol Dichotomy**: Phenol ($C_6H_5OH$) was the historic antiseptic introduced by Joseph Lister (1865). At dilute concentrations ($0.2\%$), it denatures bacterial enzymes safely on skin. At higher concentrations ($1.0\%$), it damages human epithelial tissue and is restricted to disinfecting non-living floors and drains.

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## 7.3 Industrial Glass Chemistry & Portland Cement Mechanics

### 1. Specialty Glasses

- **Crookes Glass**: Doping molten silicate glass with $CeO_2$ introduces localized electron transitions that absorb electromagnetic radiation with wavelengths below $400\ \text{nm}$ (ultraviolet spectrum) while transmitting visible light ($400–700\ \text{nm}$).
- **Pyrex (Borosilicate)**: Incorporating $B_2O_3$ reduces the coefficient of thermal expansion from $9 \times 10^{-6}\ \text{K}^{-1}$ (soda glass) to $3.3 \times 10^{-6}\ \text{K}^{-1}$. When heated unevenly, thermal expansion differentials are minimal, preventing mechanical stress cracks.

### 2. Portland Cement Hydration & Setting

- When cement is mixed with water, hydration reactions form interlocking needle-like crystals of calcium silicate hydrate (C-S-H gel) and calcium hydroxide ($Ca(OH)_2$):

$$2(3CaO \cdot SiO_2) + 6H_2O \rightarrow 3CaO \cdot 2SiO_2 \cdot 3H_2O\ (\text{C-S-H gel}) + 3Ca(OH)_2$$

- Gypsum ($CaSO_4 \cdot 2H_2O$) retards hydration of tricalcium aluminate ($C_3A$), ensuring a controlled setting profile.

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## 7.4 Agricultural Fertilizers & Soil Remediation Chemistry

### 1. Urea and Nitrogen Dynamics

- Urea ($NH_2CONH_2$, $M_w = 60\ \text{g/mol}$) contains two nitrogen atoms:

$$\%N = \frac{2 \times 14}{60} \times 100 = 46.67\% \approx 46\%$$

- **Enzymatic Soil Hydrolysis**:

$$NH_2CONH_2 + 2H_2O \xrightarrow{\text{urease}} (NH_4)_2CO_3 \rightleftharpoons 2NH_4^+ + CO_3^{2-}$$

- **Nitrification by Soil Microbes**:

$$2NH_4^+ + 3O_2 \xrightarrow{\text{Nitrosomonas}} 2NO_2^- + 4H^+ + 2H_2O$$

$$2NO_2^- + O_2 \xrightarrow{\text{Nitrobacter}} 2NO_3^-\ (\text{absorbed by roots})$$

### 2. Remediation of Problem Soils

- **Acidic Soils**: High rainfall leaches basic cations ($Ca^{2+}, Mg^{2+}, K^+$), leaving soil dominated by $H^+$ and $Al^{3+}$. Corrected by liming with $CaCO_3$ or $Ca(OH)_2$:

$$Al^{3+}\text{-Clay} + 3Ca(OH)_2 \rightarrow Ca^{2+}\text{-Clay} + Al(OH)_3 \downarrow$$

- **Alkaline / Sodic (Usar) Soils**: Arid evaporation concentrates exchangeable sodium ($Na^+$) on clay particles, causing soil swelling and poor drainage ($\text{pH} > 8.5$). Corrected with Gypsum ($CaSO_4 \cdot 2H_2O$):

$$2Na^+\text{-Clay} + CaSO_4 \rightarrow Ca^{2+}\text{-Clay} + Na_2SO_4\ (\text{soluble, flushed out by irrigation})$$

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## Bilingual Terminology -- Chemistry in Daily Life and Agriculture

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Fertiliser** (उर्वरक) | रासायनिक उर्वरक | Chemical compound supplying nutrients to plants (N, P, K) |
| **Urea** (यूरिया) | यूरिया | Most used nitrogenous fertiliser; CO(NH2)2; 46% N |
| **DAP** (डीएपी) | डाईअमोनियम फॉस्फेट | Diammonium Phosphate; N + P fertiliser |
| **Pesticide** (कीटनाशक) | कीटनाशी / कीटनाशक | Chemical that kills pests; insecticide, fungicide, herbicide |
| **DDT** (डीडीटी) | डीडीटी | Dichlorodiphenyltrichloroethane; organochlorine insecticide; banned (persistent) |
| **Insecticide** (कीटनाशी) | कीटनाशी | Kills insects; e.g., malathion, BHC, pyrethroids |
| **Herbicide** (शाकनाशी) | शाकनाशी | Kills weeds; e.g., 2,4-D; Agent Orange (2,4,5-T + 2,4-D) |
| **Fungicide** (कवकनाशी) | फफूंदनाशक | Kills fungi; e.g., Bordeaux mixture (CuSO4 + lime) |
| **Preservative** (परिरक्षक) | परिरक्षक | Food additive preventing spoilage; salt, sugar, vinegar, sodium benzoate |
| **Food Additive** (खाद्य योज्य) | खाद्य योजक | Substance added to food for colour, flavour, preservation, nutrition |
| **Dye** (रंजक) | रंजक | Coloured substance; synthetic (aniline dyes) or natural (indigo, turmeric) |
| **Soap** (साबुन) | साबुन | Sodium/potassium salt of fatty acids; biodegradable cleansing agent |
| **Detergent** (अपमार्जक) | अपमार्जक | Synthetic cleaner; sulphonate/sulphate; works in hard water |
| **Antiseptic** (रोगाणुनाशक) | रोगाणुनाशक | Kills microorganisms on living tissue; dettol, savlon, iodine |
| **Disinfectant** (विसंक्रामक) | विसंक्रामक | Kills microorganisms on inanimate surfaces; bleach, phenol |
| **Drug / Medicine** (औषधि) | औषधि / दवा | Chemical compound used to treat, cure, or prevent disease |

## One-Liner Revision -- Chemistry in Daily Life and Agriculture

| # | Fast Fact |
|---|---|
| 1 | Urea (CO(NH2)2) = most widely used nitrogen fertiliser; 46% N; water-soluble |
| 2 | DAP = Diammonium Phosphate; provides both N and P to crops |

| # | Fast Fact |
|---|---|
| 3 | NPK fertilisers = provide Nitrogen, Phosphorus, Potassium (potash) |
| 4 | Bordeaux mixture = CuSO4 + Ca(OH)2; first effective fungicide; grape vines, potato blight |
| 5 | DDT = banned insecticide (persistent organic pollutant); biomagnification in food chain |
| 6 | Biomagnification = increasing concentration of persistent toxin up the food chain |
| 7 | BHC/BHC (Benzene Hexachloride) = lindane; insecticide; restricted use |
| 8 | Common antiseptics: Dettol (chloroxylenol), Savlon (chlorhexidine), Iodine (tincture) |
| 9 | Common disinfectants: bleaching powder, phenol (carbolic acid), formaldehyde |
| 10 | Paracetamol = analgesic (pain reliever) + antipyretic (fever reducer); most common OTC drug |
| 11 | Aspirin (acetylsalicylic acid) = analgesic, antipyretic, anti-inflammatory; anti-platelet |
| 12 | Antacids neutralise excess HCl: Mg(OH)2 (milk of magnesia), Al(OH)3, NaHCO3 |
| 13 | Tranquilizers = CNS depressants; reduce anxiety; e.g., diazepam (Valium), barbiturates |
| 14 | Antibiotics kill bacteria; NOT effective against viruses |
| 15 | Antihistamines = counter histamine (allergic reaction); e.g., cetirizine, diphenhydramine |
| 16 | Soap = biodegradable; forms scum in hard water (Ca/Mg stearate precipitate) |
| 17 | Detergent = synthetic (non-soap); does NOT form scum in hard water |
| 18 | Food preservatives: salt (NaCl), sugar, vinegar (acetic acid), sodium benzoate, SO2 |
| 19 | Food colours: Tartrazine (E102, yellow), Carmoisine (red) are synthetic; saffron, turmeric are natural |
| 20 | Baking powder = baking soda (NaHCO3) + cream of tartar (tartaric acid); releases CO2 |
| 21 | Vinegar = 4-8% acetic acid (CH3COOH); food preservative and condiment |
| 22 | Chlorination of water = Cl2 or bleaching powder kills pathogens; residual Cl = 0.2-0.5 ppm |
| 23 | Fluoridation of water = NaF added; 0.7-1.2 ppm F- prevents tooth decay |
| 24 | Iodised salt = NaCl + KIO3 (potassium iodate); prevents iodine deficiency/goitre |
| 25 | Organic farming = no synthetic pesticides/fertilisers; uses compost, biopesticides |
| 26 | Green Chemistry = design of chemical products that minimise environmental hazards |
| 27 | Food packaging: BPA (Bisphenol A) in polycarbonate plastics = endocrine disruptor |
| 28 | MSG (Monosodium glutamate) = flavour enhancer; "Umami" taste |
| 29 | Aspartame = artificial sweetener; ~200x sweeter than sugar; 0 calories |
| 30 | Saccharin = oldest artificial sweetener (1879); 300-500x sweeter; no calories |